

Irrational inequalities

Homework

№1. Solve the following equations by method of a careful look:

1. $\sqrt{x-8} - \sqrt{x+2} < 0;$

2. $\sqrt{9-x^2} < -|x^2+4x+3|;$

3. $\sqrt{x-1} > x+1;$

4. $\frac{\sqrt[4]{x^4+16} + \sqrt{x^3+8}}{2-x} < 0;$

5. $\sqrt{x+1} + \sqrt{x-2} \geq \sqrt{2-x}.$

№2. Solve the following equations:

1. $\sqrt{5-x^2} > x-1;$

2. $x^2 \geq x \left(2 + \sqrt{12-2x-x^2} \right);$

3. $1-x < \sqrt{x^2-2x};$

4. $\sqrt{3x-x^2} < 4-x;$

5. $\sqrt{x}-3 \leq \frac{2}{\sqrt{x-2}};$

6. $\frac{\sqrt{2x-1}}{x-2} < 1;$

7. $\frac{1-\sqrt{1-4x^2}}{x} < 3.$

№3. Solve the following equations:

1. $\frac{3}{\sqrt{2-x}} - \sqrt{2-x} < 2;$

2. $\sqrt{2x-x^2} \leq x-2.$

№4. Solve an equation:

$$\sqrt{4-x^2} + \frac{|x|}{x} \geq 0$$